MIL-P-466A
24 APRIL 1951
SUPERSEDING
JAN-P-466
28 April 1947

MILITARY SPECIFICATION

POWDER, PHOTOFLASH

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 Scope.—This photoflash powder is intended for use in photoflash ammunition, miscellaneous photographic uses, and for special effects.

1.2 Classification.—This specification covers the following types and classes of photoflash powder (see 6.1):

Type I —Mixed base of aluminum and magnesium powders
Class A—Potassium perchlorate oxidizing agent.

Type II —Magnesium-aluminum alloy powder base
Class A—Potassium perchlorate oxidizing agent.
Class B—Barium nitrate oxidizing agent.

Type III—Aluminum powder base
Class A—Mixed oxidizing agent of potassium perchlorate and barium nitrate.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 Specifications.—The following specifications and specifications referenced on drawings in paragraph 2.3 of the issue in effect on date of invitation for bids form a part of this specification.

FEDERAL SPECIFICATION

RR-S-366 —Sieves, Standard, Testing.

MILITARY SPECIFICATIONS

JAN-A-289—Aluminum Powder, Flaked, Grained and Atomized (for use in ammunition).
JAN-B-162—Barium Nitrate.
JAN-B-217—Potassium Perchlorate.
JAN-M-382—Magnesium Powder (for use in ammunition).
JAN-M-454—Magnesium-Aluminum Alloy, Powdered.
MIL-G-2550—General Specification for Ammunition Except Small Arms Ammunition.

U. S. ARMY SPECIFICATIONS

100-2 —Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.

(*Army.*—Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of Federal, Military, and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Sup-

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ply Depot, Scotia 2, N. Y. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 **Other publications.** — The following publications of the issue in effect on date of invitation for bids form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATIONS

Navy Shipment Marking Handbook

(Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C. except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, N. Y.)

INTERSTATE COMMERCE COMMISSION REGULATIONS

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

(Information as to the availability of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., may be obtained from the Interstate Commerce Commission, Washington 25, D. C.)

2.3 **Drawings.**—The following drawing, of the issue in effect on date of invitation for bids, forms a part of this specification.

ORDNANCE CORPS DRAWING

76–4–43—Container, Black Powder, Storage and Shipping.

(Copies of drawings should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 **Material.**—The raw material used in the manufacture of photoflash powder shall conform with the requirements of the applicable specifications listed in paragraph 2.1.

3.1.1 *Magnesium.*—Magnesium shall conform to the requirements for type I, grade A of Specification JAN–M–382. There shall be no requirement with respect to apparent density and the granulation shall be as follows using U. S. Standard sieves conforming to Federal Specification RR–S–366:

Through No. 80 sieve 100 percent, min.
Through No. 200 sieve 60 percent, min.

3.1.2 *Aluminum.*

3.1.2.1 *Type I photoflash powder.*—Aluminum shall conform to the requirements for type A of Specification JAN–A–289 except that the free metallic aluminum content shall be not less than 95.0 percent and the granulation shall be as follows:

Through No. 80 sieve 100 percent, min.
Through No. 200 sieve 60 percent, min.

3.1.2.2 *Type III photoflash powder.*—Aluminum shall conform to the requirements for type C of Specification JAN–A–289 except that the average particle size shall be 20 ± 5 microns.

3.1.3 *Magnesium-aluminum alloy.* — Magnesium-aluminum alloy powder shall conform to the requirements for type A of Specification JAN–M–454 except that the granulation shall be as follows:

Through No. 80 sieve 100 percent, min.
Through No. 200 sieve 60 percent, min.

3.1.4 *Potassium perchlorate.*

3.1.4.1 *Type I and Type II photoflash powders.*—Potassium perchlorate shall conform to the requirements for grade A of Specification JAN–P–217.

3.1.4.2 *Type III photoflash powder.*—Potassium perchlorate shall conform to the requirements for grade A of Specification JAN–P–217 except that the average particle size shall be 17 ± 2 microns.

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3.1.5 *Barium nitrate.*

3.1.5.1 *Type II photoflash powder.*—Barium nitrate shall conform to the requirements for class B of Specification JAN–B–162 except that the granulation shall be as follows:

Through No. 100 sieve 100 percent, min.
Through No. 200 sieve 95 percent, min.

3.1.5.2 *Type III photoflash powder.*—Barium nitrate shall conform to the requirements for class D of Specification JAN–B–162 except that the average particle size shall be 75 ± 25 microns.

3.2 **Chemical composition.**—The chemical composition of photoflash powder shall conform to the requirements shown in table I.

3.3 **Moisture.**

3.3.1 *Types I and II.*—Maximum, 0.2 percent.

3.3.2 *Type III.*—Maximum, 0.1 percent.

3.4 **Grease, Types I and II, only.**—Maximum, 1.0 percent.

3.5 **Granulation.**

3.5.1 *Types I and II.*

Through No. 80 sieve 100 percent, min.
Through No. 200 sieve 60 percent, min.

3.5.2 *Type III.*—Granulation shall be as indicated in applicable paragraphs of 3.1.

3.6 **Workmanship.** — The finished powder shall be a uniform mixture free from lumps or a tendency to form lumps.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Lot.** — A lot shall consist of 25,000 pounds, maximum.

4.2 **Sampling.**—Ten percent, but in no case more than 10 or less than 3, of the containers shall be selected by the inspector so as to be representative of the lot. If there are fewer than 3 containers in the lot, all the containers shall be sampled. The material shall be mixed to a uniform consistency throughout and approximately 8 ounces of it shall be taken from each selected container. Each of these primary samples shall be placed in an airtight container and labeled so that the container from which it was taken can be identified. A composite sample of approximately 1 pound shall be made from equal portions of the primary samples. The composite sample shall be thoroughly mixed and placed in an airtight container labeled to show the name of the material, manufacturer, plant, contract or order number, lot number and lot size. All acceptance tests shall be made on the composite sample. However, if it becomes apparent during sampling that the lot is not uniform, the inspector may require that any primary sample be tested for compliance with the requirements of the specification. All primary samples shall be held for possible future examination should

TABLE I.—*Chemical composition*

	Type I	*Type II*		*Type III*
	Class A	*Class A*	*Class B*	*Class A*
	Percent	*Percent*	*Percent*	*Percent*
Magnesium	34.0 ± 1.0			
Aluminum	26.0 ± 1.0			40.0 ± 1.0
Magnesium-aluminum alloy		60.0 ± 1.0	44.5 ± 1.0	
Potassium perchlorate	40.0 ± 1.0	40.0 ± 1.0		30.0 ± 1.0
Barium nitrate			55.5 ± 1.0	30.0 ± 1.0

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the composite sample fail to meet the requirements.

4.3 Inspection.

4.3.1 *Army.*—Inspection shall be made in accordance with Specification MIL–G–2550 and shall be made at the point of delivery unless otherwise specified in the contract or order.

4.3.2 *Navy.*—Inspection shall be made at the point of delivery unless otherwise specified in the contract or order. The general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.4 Moisture.—Transfer a weighed portion of approximately 5 gm of the sample to a tared weighing bottle. Dry the bottle and contents at 100° to 105° C. for 3 hours, cool in a desiccator, and weigh. Calculate the loss in weight as percent moisture.

4.5 Composition.

4.5.1 *Type I, Class A.*

4.5.1.1 *Potassium perchlorate.*—Transfer a weighed portion of approximately 2 gm. of the sample to a 250-ml. beaker and surround this with ice water. Add 100 ml. of distilled water at a temperature of 0° to 5° C. Stir the mixture vigorously for 8 minutes scraping down the inner wall of the beaker. Transfer the residue to a tared Gooch crucible and wash thoroughly with distilled water at 0° to 5° C. Wash the residue with alcohol, then with ether, dry the crucible and contents at 100° to 105° C. for 1 hour, cool in a desiccator and weigh. Calculate the loss in weight of the sample as percent potassium perchlorate on a moisture-free basis.

4.5.1.2 *Magnesium.*—Extract the residue from the potassium perchlorate determination with 5 percent acetic acid until all magnesium is dissolved, as indicated by no evolution of gas when the residue is treated with fresh portions of the acetic acid solution. Wash the undissolved residue with distilled water, rinse with alcohol and then with ether, heat the crucible and contents for an hours at 60° C., cool in a desiccator, and weigh. Calculate the loss in weight as percent magnesium on a moisture-free basis.

4.5.1.3 *Aluminum.*—Calculate the weight of undissolved residue in the crucible (see 4.5.1.2) as percent aluminum on a moisture-free basis.

4.5.2 *Type II, Class A.*

4.5.2.1 *Potassium perchlorate.* — Use the same procedure as prescribed in paragraph 4.5.1.1.

4.5.2.2 *Magnesium-aluminum alloy.*—Calculate the weight of the undissolved residue in the crucible remaining from the determination of the potassium perchlorate as percent magnesium-aluminum alloy on a moisture-free basis.

4.5.3 *Type II, Class B.*

4.5.3.1 *Barium nitrate.*—Use the same procedure as specified for potassium perchlorate in paragraph 4.5.1.1.

4.5.3.2 *Magnesium-aluminum alloy.* — Use the same procedure as prescribed in paragraph 4.5.2.2.

4.5.4 *Type III, Class A.*

4.5.4.1 *Aluminum.*—Transfer an accurately weighed portion of approximately two grams of the sample to a tared medium porosity sintered glass crucible. Extract the sample with ten 25-ml. portions of distilled water at a temperature of 0° to 5° C., using a filtering flask and suction (see 6.2). Stir the contents of the crucible with each portion of water for 1 minute before applying suction.

Transfer the extract quantitatively to a 400 ml. beaker using approximately 50 ml. of wash water and reserve the extract and washings for the barium nitrate determina-

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tion. Rinse the undissolved residue with alcohol and ether, and aspirate until no further solvent odors are apparent. Place the crucible and contents in a drying oven at 100° to 105° C. for 1 hour, cool in a desiccator and weigh. Calculate the increase in weight of the crucible to percent aluminum powder on a moisture-free basis as follows:

$$\text{Percentage of aluminum powder} = \frac{100A}{W\ (1-M)}$$

where:

A = increase in weight of filtering crucible
W = weight of sample
M = percent moisture expressed as a decimal fraction.

4.5.4.2 *Barium nitrate.*—To the extract reserved during the aluminum powder determination, add 1 ml. of 12N hydrochloric acid. Heat to boiling on a hot plate and add slowly from a 50-ml. pipette 50 ml. of 3N (1:8) sulfuric acid. Allow the mixture to boil gently for five minutes and finally to digest for at least two hours on a steam bath. Collect the precipitated barium sulfate in a tared Gooch crucible which has been previously heated in a muffle furnace at approximately 750° C. for 30 minutes. Wash the precipitate ten times with 25-ml. portions of hot water, dry in an oven at 100° ± 5° C. for 1 hour and in a muffle furnace at approximately 750° C. for 30 minutes, cool in a desiccator and weigh. Calculate the percentage of barium nitrate in the sample on a moisture-free basis as follows:

$$\text{Percentage of barium nitrate} = \frac{111.98A}{W\ (1-M)}$$

Where:

A = increase in weight of Gooch crucible
W = weight of sample
M = percent moisture expressed as a decimal fraction.

4.5.4.3 *Potassium perchlorate.*—Calculate the percentage of potassium perchlorate in the sample by deducting the percentages of barium nitrate and aluminum powder found from 100.

4.6 Grease, Types I and II only.—Transfer a weighed portion of approximately 10 gm. of the sample to a beaker, add 50 ml. of anhydrous ether, and stir. After a few minutes, filter the ether into a tared 250-ml. beaker. Wash the undissolved solids with two additional 50-ml. portions of anhydrous ether filtering each into the same beaker as the first. Evaporate the ether from the beaker by heating or by a current of dry air. Heat the beaker and residue at 60° C. for one hour, cool in a desiccator, and weigh. Calculate the increase in weight as percent grease on a moisture-free basis.

4.7 Granulation.

4.7.1 *Types I and II.*—Place a weighed portion of approximately 10 gm. of the sample of the nest of sieves specified in paragraph 3.5 properly superimposed and assembled with a cover and a bottom pan. Shake for 10 minutes by means of a mechanical shaker which imparts to the sieves a rotary motion and tapping action of uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the material remaining on each sieve and calculate the percentage passing through each sieve.

4.7.2 *Type III.*—The particle size of the individual ingredients shall be determined by use of a Fisher Sub-Sieve Sizer or equal.

4.8 Resubmission and retest.—If the composite sample or any primary sample subjected to test fails to pass the test, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on samples from any or all of the containers in the lot at no expense to the Government. The contractor may then remove the defective portions of the lot and resubmit the lot for acceptance, provided complete replacement of the defective portions can be made to the satisfaction of the inspector. The resubmitted lot shall be accepted provided that new samples selected in accordance with paragraph 4.2 pass all of the tests required by the specification.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.—Unless otherwise specified, packaging and packing shall conform to the requirements for shipment of inflammable solids and oxidizing materials as specified in Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc. In addition, packing containers shall conform to Army Ordnance Drawing 76–4–43 or a commercial container of equal quality. A paper bag liner shall be used. A gasket which completely covers the opening shall be provided to prevent any possibility of loose powder becoming lodged in the screw heads. The container shall be painted black.

5.2 Marking.—In addition to any special marking required by the contract or order and marking to insure safe handling as required by the Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., shipments for the Army shall be marked in accordance with the requirements of the U. S. Army Specification 100–2; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

6. NOTES

6.1 Ordering data. — Invitations for bids and contracts or orders should contain the following:

a. Title, number, and date of the specification.
b. Type and class required. (See par. 1.2.)

6.2 A Fisher Filtrator, Fisher Cat. No. 9–788, has been found satisfactory for this purpose.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army—Ordnance Corps

Other interest:
Army—S
Navy—OS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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